**S475/1**
**SUBSIDIARY MATHEMATICS**
**Paper 1**
**July/Aug. 2026**
**$2\frac{1}{2}$ hours**

**KAGADI JOINT MOCK EXAMINATIONS CONSULT**

**Uganda Advanced Certificate of Education**

**SUBSIDIARY MATHEMATICS**

**Paper 1**

2 hours 30 minutes

**INSTRUCTIONS TO CANDIDATES:**

*This paper consists of two Sections:* **A** *and* **B**. *It has* **six** *examination items.*

***Section*** **A** *has* **two compulsory** *items.*

***Section*** **B** *has two parts:* **I** *and* **II**. *Respond to* **one** *item from each part.*

*Respond to* ***four*** *items in all.*

*Any additional item responded to* will **not** *be scored.*

*All responses must be written in the answer booklet(s) provided.*

*Begin each part of the item on a fresh page.*

*Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

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## SECTION A

*Respond to* **both** *items in this section.*

### Item 1

During the preparations for the commissioning of a newly completed sports complex, the engineering team is installing three communication masts to ensure uninterrupted transmission during national sports events.

The control station is represented by the point $O(0,0)$ on a Cartesian plane. Two communication masts have already been erected at $A(56, 42)$ and $B(-48, 64)$.

The technical regulation require that a third mast shall only be installed if the angle subtended by $OA$ and $OB$ exceeds $60^0$.

For maximum signal reception, the transmitting antenna is to be inclined at an angle of $\theta$ to the horizontal satisfying

$$2\sin\theta + \sqrt{3}\cos\theta = 2$$

The engineering team requires mathematical guidance before authorizing the installation.

**Task:**

Determine:

(a) whether the installation satisfies the condition for erecting the third mast, giving appropriate justification.

(b) angle at which the antenna should be inclined and advise the engineering team.

### Item 2

**Mpandwe Irrigation Authority** monitors the volume of water stored in a reservoir supplying several large farms. Records indicate that the volume of stored water, $V$ cubic metres, after $x$ months is represented by

$$V(x) = -400x^2 + 4800x + 24000.$$

The authority intends to present a technical report showing how the quantity of stored water changes with time before planning the next phase of expansion.
At the same time, engineers discover leakage from one of the storage tanks. Investigations reveal that the rate at which the water leaves the tank is directly proportional to the amount of water remaining in the tank. Initially the tank contains 12000 cubic meters of water and the proportionality constant is 0.05 per month.

The authority requires a comprehensive mathematical analysis before deciding on the appropriate intervention.

**Task:**

(a) Carry out a complete analysis of the reservoir model, including an appropriate graph and interpretation of the important features.

(b) Determine the amount of water remaining in the tank after 6 months and advise the authority accordingly.

## SECTION B

### Part I

*Respond to* **one** *item from each part.*

**Item 3**

A young entrepreneur in one of the cities in Uganda is reviewing her new dairy processing business. She needs help to decide whether to expand her factory space based on her weekly yogurt sales to schools and supermarkets. Her weekly profit, ( in thousands of shillings) is modelled by: $P(x) = -2x^2 + 20x - 32$, were $x$ is the number of hundreds litres of yogurt made. She will only proceed with the expansion if her maximum possible weekly profit exceeds Shs 15,000.
For buy new packing machine, she saves Shs 150,000 in the first month and increases this amount by Shs 30,000 every month. The machine she needs cost Shs 3,000,000 but she does not know whether will get that money by the end of 10 months.

At the same time, she has a separate investment fund growing by the rule: $V(n) = 200{,}000 \times (1.05)^n$, after $n$ months. Furthermore, she needs to choose 3 packaging workers from 8 job seekers consisting of 5 women and 3 men. She will buy a new packing machine the exact month her investment fund exceeds Shs 500,000, and she will only finalise her hiring if there are more than 50 valid ways to form a team that includes at least one woman.

**Task:**

Help the young entrepreneur resolve her challenges.

**Item 4**

A bakery in a certain town is expanding its bread production to meet the increasing demand and has been given a contract to produce and supply 690 trays of bread to a supermarket within 10 days.

To fulfil the contract, the bakery has started by producing 6 trays of bread on the first day, will produce 9 trays on the second day, 12 trays on the third day and the bread production will keep increasing by the same constant number of trays each subsequent day.

The bakery manager wants to determine how long it will take to fulfil the contract if production continues to increase in the same pattern.

The bakery produces 6 different types of bread. A nearby shop wants to order for 3 different types of bread, however, the shopkeeper does not know the different possible number of selections to choose from if the order in which the types are chosen does not matter.

**Task**:

(a) (i) Assess whether the production pattern will enable the manager to fulfil the contract.

(ii) Recommend an alternative production pattern for the manager to fulfil the contract, assuming that 6 trays are produced on the first day.

(b) Determine the number of possible bread type selections the shopkeeper can choose from.

## Part II

*Respond to* **one** *item from each part.*

**Item 5**

The head teacher of a boarding school is evaluating the school's feeding program for next term amidst changing market conditions. She collected price data for four foundational food groups used in the school kitchen for Term 1(*Base Period*) and Term 2 (*Current Period*). She needs help to know if the cost of living has increased in order to enhance on the budget for next term.

| Food Group | Term 1 Price (UGX) | Term 2 Price (UGX) | Weight |
|---|---|---|---|
| Maize Flour | 2,500 | 3,200 | 120 |
| Beans | 3,500 | 4,000 | 80 |
| Rice | 4,500 | 5,200 | 50 |
| Sugar | 4,000 | 4,800 | 30 |

The school clinic distributes glucose packets ($x$) to weak students during sports days. The table below shows the chance of a student getting each number of packets.

| Number of packets($x$) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Chance $P(X = x)$ | $c$ | $2c$ | $3c$ | 0.4 |

The head teacher considers the distribution plan fair if the variation in the number of glucose packets distributed is within an acceptable limit of 2 packets.

She wishes to use appropriate statistical evidence to determine whether the plan should be retained.

**Task:**

Help the head teacher to address her concerns.

**Item 6**

A nursery school in Ntinda is focused on developing both the reading and writing skills for their toddlers in preparation for primary level. Mr Charles wants to understand whether there is a relationship between his pupils' reading and writing abilities. She believes that pupils who perform well in reading may also perform well in writing, but she wants to use data to confirm this before designing a combined reading-writing support program. She records the following scores for nine pupils:

| **Pupil** | A | B | C | D | E | F | G | H | H |
|---|---|---|---|---|---|---|---|---|---|
| Reading score | 62 | 81 | 74 | 20 | 36 | 41 | 25 | 91 | 55 |
| Writing Score | 78 | 71 | 86 | 57 | 62 | 52 | 37 | 75 | 72 |

In the same class, due to changes in weather patterns, the head teacher decided to motivate the teachers by giving them some fruits. Charles and Betty are some of the teachers in this school. The head teacher came with a box with 5 green apples and 7 red apples, every teacher was supposed to pick two apples one after the other without looking into the box.
The teachers wanted to know the chance of selecting the both red apples

**Task:**

Help the teachers to determine the:

(a) Relationship between the reading and writing scores.

(b) The chance that;

(i) madam Betty will pick both red apples.

(ii) Mr Charles will pick apples of different colours.



**END**